

INDIAN INSTITUTE OF TECHNOLOGY,
MADRAS

BUSINESS DATA MANAGEMENT CAPSTONE PROJECT
FINAL SUBMISSION REPORT



Title: Optimizing Data Management for Efficient Operations in a
Soft Drink Wholesale store.

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1: EXECUTIVE SUMMARY

Title: Optimizing Data Management for Efficient Operations in a Soft Drink

Wholesale Store

This report presents a complete analysis of the challenges faced by SHIVA ENTERPRISES, a local wholesale dealer of cold drinks based in Ghazipur, Uttar Pradesh. The organisation has faced difficulties related to tough market competition, seasonal sale variations, pricing strategies, etc. This analysis focuses on providing insights and various recommendations to face these challenges and promote business.

The analysis process includes the utilization of data provided by organisation including sales data, profit and loss statements for the month of July'23 to Sept'23. Despite the available data didn't contain information about category and parent company, some necessary changes were applied to get meaningful insights.

Our data analysis discovered valuable information about demand patterns, customer preferences and sales pattern. Seasonal demand variations were evident, emphasizing the importance of flexible inventory management. Key findings indicates that high demand of products was in month of summer specifically and during festival and marriage seasons. The analysis also shows the need of improved sales forecasting, pricing optimization, diversification of sales channels to makes sales consistent.

By implying advanced analytical tools, including demand forecasting, inventory optimization, this report proposes actionable recommendations. Recommendations includes tailored inventory management strategies, optimizing pricing strategies to navigate pricing wars, and targeted marketing efforts.

By implying the insights and recommendations provided in this analysis, the business can overcome from the challenges faced by them, strengthen their market presence and achieve sustainable growth. The report underscores the importance of data usage to make decisions and crafting strategic planning to achieve success in fiercely competitive market.

2: DETAILED EXPLANATIONS OF ANALYSIS PROCESS/METHOD

Click on: [Dataset](#)

Introduction to the Analysis Process:

The analysis in this project aimed to tackle the challenges faced by Shiva Enterprises, a local Cold Drink wholesale store. It focuses on understanding sales and purchase patterns and tracking sales trends over time. The goal was to use this analysis to offer practical recommendations for overcoming challenges and fostering business expansion. Analysing sales and purchase data is vital for grasping the factors influencing a business's performance. It helps uncover trends, patterns, and areas for enhancement. These insights empower us to make informed decisions, enhance inventory control, and devise strategies for boosting sales and profitability. In essence, the analysis process is crucial for identifying opportunities, managing risks, and improving the competitiveness of the business. By using data-driven insights, we can address challenges proactively and promote sustainable growth in a competitive market.

Data Preparation:

After having a few rounds of discussion with the owner, we got the Raw data which required modifications to get meaningful insights for analysis and decision making.

	A	B	C	D	E	F	G	H	I	J
1	DATE	CUSTOMER NAME	COMPANY	PRODUCT	PACKAGE CATEGORY	PURCHASE RATE	SELLING RATE	QUANTITY(BOXES)	MODE OF PAYMENT	LOCATION
2	01-07-2023	AMARNATH	COCA-COLA	SPRITE	1.25L	640	690	3	UPI	NAVAPURA
3	01-07-2023	AMARNATH	COCA-COLA	SPRITE	2.25L	750	810	2	UPI	NAVAPURA
4	01-07-2023	AMARNATH	PARLE AGRO	BAILEY	250ML	120	156	12	UPI	NAVAPURA
5	01-07-2023	AMARNATH	PEPSICO	PEPSI	750ML	440	570	4	UPI	NAVAPURA
6	01-07-2023	AKRAM	COCA-COLA	SPRITE	750ML	510	570	3	UPI	BALAPUR
7	01-07-2023	AKRAM	COCA-COLA	THUBS UP	2L	660	815	2	UPI	BALAPUR
8	01-07-2023	SUNIL	COCA-COLA	SPRITE	750ML	510	570	3	UPI	SHERPUR

Fig: Raw data

The raw data provides information about the date of transaction, customer name, company name and name of product, package category, purchasing rate, selling rate, quantities of product sold, mode of payment and location of customer as well. This schema helps in analysis and decision making related to pricing and discount strategies, inventory management and market share.

While processing data during data arrangement phase, the raw data underwent several transformations to extract key insights related to the store's financial and operational

performance. One significant transformation involved adding two new columns: "Total" and "Daily Profit," aimed at enhancing the dataset's analytical capabilities.

The "Total" column was calculated by multiplying the Selling Rate of a product by the Quantity of items sold, representing the total price of the product. On the other hand, the "Daily Profit" column was derived by subtracting the Purchase Rate from the Selling Rate and then multiplying the result by the Quantity of the product sold. This column provides crucial information on the daily profit generated by the store, aiding in understanding profitability trends, financial performance over time, and making informed decisions regarding profit fluctuations.

Additionally, columns such as "Day," "Month," and "Company" were added to the dataset. These columns enable the extraction of day and month-wise information regarding customer engagement, demand, sales, and profitability. The "Company" column provides insights into the parent company of products, facilitating the analysis of demand and sales trends for products from specific companies.

1	DATE	DAY	MONTH	CUSTOMER NAME	COMPANY	PRODUCT	CATEGORY	PACKAGE CATEGORY	PURCHASE RATE	SELLING RATE	QUANTITY(BOXES)	MODE OF PAYMENT	TOTAL	DAILY PROFIT	LOCATION
2	01-07-2023	Saturday	July	AMARNATH	COCA-COLA	SPRITE	SOFT DRINKS	1.25L	640	690	3	UPI	2070	150	NAVAPURA
3	01-07-2023	Saturday	July	AMARNATH	COCA-COLA	SPRITE	SOFT DRINKS	2.25L	750	810	2	UPI	1620	120	NAVAPURA
4	01-07-2023	Saturday	July	AMARNATH	PARLE AGRO	BAILEY	WATER	250ML	120	156	12	UPI	1872	432	NAVAPURA
5	01-07-2023	Saturday	July	AMARNATH	PEPSICO	PEPSI	SOFT DRINKS	750ML	440	570	4	UPI	2280	520	NAVAPURA
6	01-07-2023	Saturday	July	AKRAM	COCA-COLA	SPRITE	SOFT DRINKS	750ML	510	570	3	UPI	1710	180	BALAPUR
7	01-07-2023	Saturday	July	AKRAM	COCA-COLA	THUBS UP	SOFT DRINKS	2L	660	815	2	UPI	1630	310	BALAPUR

Fig: Processed Data

By enhancing the dataset with these additional columns, Shiva Enterprises can access a more thorough and precise dataset for examination. This refined data will empower the company to conduct detailed financial assessments, assess profitability, establish market rates, and make well-informed choices regarding financial management, pricing strategies, and market expansion initiatives.

Sales Analysis:

To analyze the sales and purchase data effectively, various steps were undertaken to extract insights and comprehend the trends. The primary emphasis was on summarizing the data by month to observe patterns and facilitate informed decision-making. This process involved creating pivot tables and generating graphical representations of the data.

The initial step involved ensuring that the months were ordered correctly for particular year. This was accomplished by adding custom lists in the advanced settings of Excel. These custom lists enabled the arrangement of months in the desired order, thereby ensuring accurate chronological representation of the data

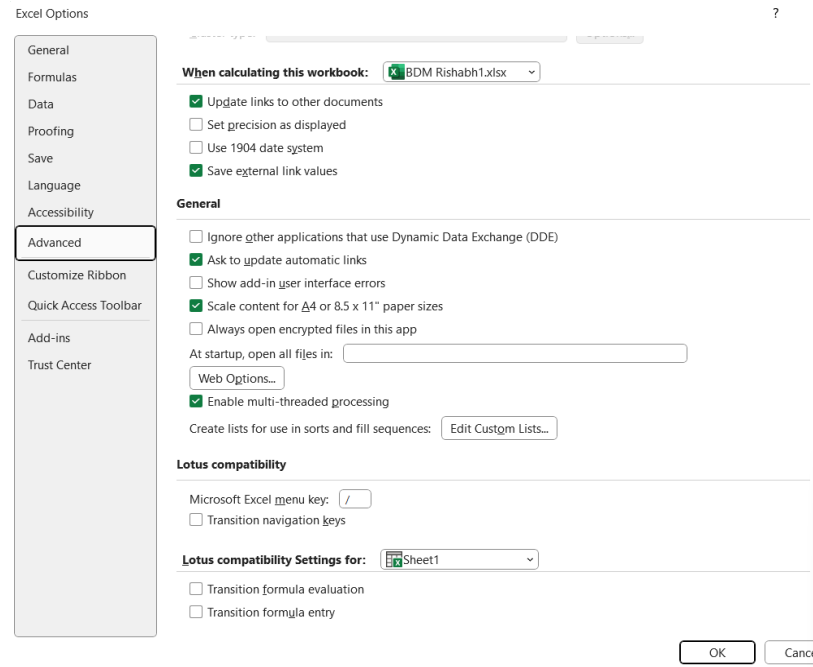


Figure: Options > Advanced > Edit Custom Lists

Following the correct ordering of months, pivot tables were constructed to summarize the sales and purchase data on monthly basis. These tables provided a structured overview of total sales amounts for each month, facilitating an easy analysis of trends over time.

To visually represent the data, bar charts were generated for sales data across each financial year. These charts offered a clear depiction of sales amounts for each month, aiding in the swift identification of trends and patterns. The bar charts were instrumental in highlighting months with notably high sales volumes.

Moreover, Pareto charts were employed to identify the top-selling month. These charts illustrated the contribution of each month to the overall sales amount, helping pinpoint months with the most significant impact on overall performance.

By analysing the sales data using pivot tables and graphical representations, key insights were gained into the performance and trends of the business. This information allowed for informed decision-making and the development of strategies to address challenges and drive growth.

3: Results and Findings

➤ 3.1 Company Products Analysis



Figure:1: Coca-Cola



Figure:2: Dabur



Figure:3: PepsiCo



Figure:4: Parle Agro



Figure:5: Bisleri

To analyse the product types based on the provided data, we'll assess their performance using three key metrics: quantities sold, total sales, and daily profits. This analysis aims to identify the most successful product categories based on their sales performance.

COMPANY	SUM of QUANTITY(BOXES)	SUM of TOTAL	SUM of DAILY PROFIT
BISLERI	359	₹59610	₹13050
COCA-COLA	1864	₹748360	₹106150
DABUR	214	₹304680	₹43220
PARLE AGRO	414	₹114217	₹20917
PEPSICO	405	₹241525	₹51810
Grand Total	3256	₹1468392	₹235147

Bisleri: Sold 359 boxes, generating a total sales revenue of ₹59,610 with a daily profit of ₹13,050. The company's sales performance indicates a consistent demand for its products, contributing significantly to its profitability.

Coca-Cola: Recorded the highest quantity sold at 1,864 boxes, resulting in total sales of ₹748,360 and a daily profit of ₹106,150. Coca-Cola's strong sales figures highlight its popularity and strong market presence, leading to substantial profitability.

Dabur: Sold 214 boxes, generating total sales of ₹304,680 and a daily profit of ₹43,220. Despite lower quantities sold compared to other companies, Dabur achieved a considerable daily profit, indicating efficient cost management and pricing strategies.

Parle Agro: Sold 414 boxes, resulting in total sales of ₹114,217 and a daily profit of ₹20,917. Parle Agro's sales performance demonstrates a stable market position, contributing positively to its daily profitability.

PepsiCo: Sold 405 boxes, generating total sales of ₹241,525 and a daily profit of ₹51,810. PepsiCo's sales figures reflect a strong market presence and effective sales strategies, leading to substantial profitability.

Overall: The analysis shows that Coca-Cola has the highest sales volume and daily profit among the companies, indicating its strong market position and profitability. Bisleri, Dabur, Parle Agro, and PepsiCo also show solid sales performance, contributing positively to their respective profitability.

➤ 3.2 Sales and Profit pareto Analysis:

The Pareto analysis was conducted to identify the distribution of total sales and profit amounts across different companies. The Pareto chart below displays the cumulative percentage of sales and profit amounts:

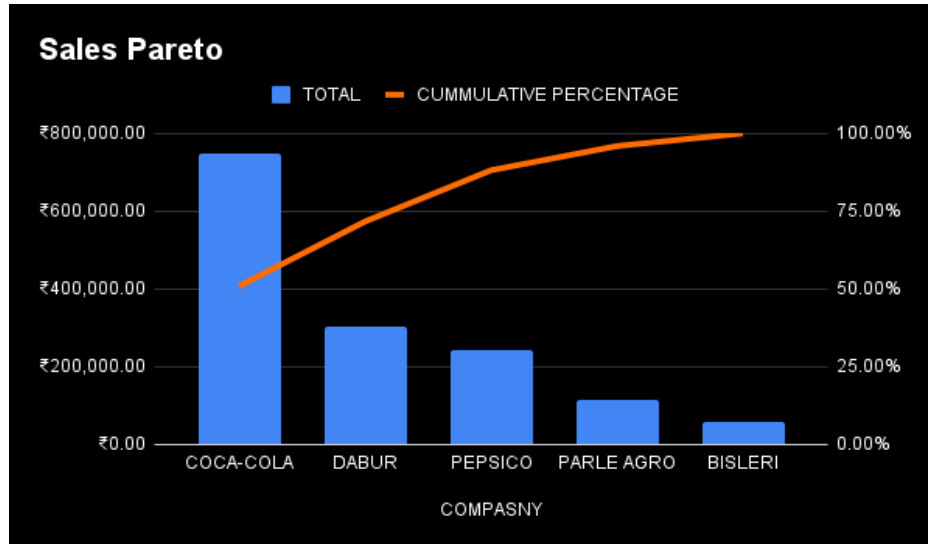


Figure:6

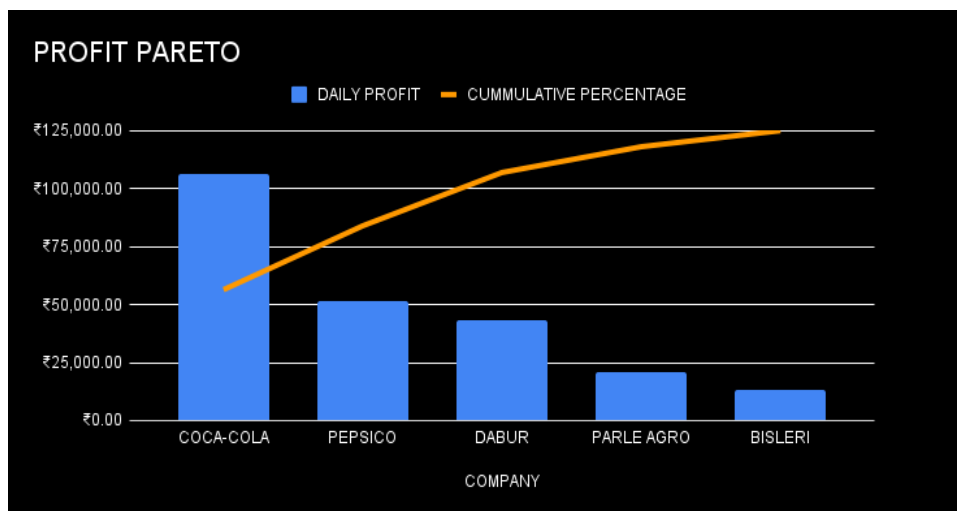


Figure:7:

Key findings from the Pareto analysis include:

Sales Pareto: This analysis revealed the distribution of sales amounts across different companies. This insight is valuable for identifying the companies that contribute the most to overall sales. Such information can guide decision-making and resource allocation for sales and marketing strategies, helping owner to focus on their efforts in peak months to maximize revenue.

Profit Pareto: The Profit Pareto analysis reveals the distribution of profits across different companies. It helps identify the companies that contribute the most to overall profits, allowing businesses to focus on maximizing profitability. By understanding which company's products are the most profitable, owner can allocate resources more effectively, prioritize marketing efforts, and make strategic decisions to enhance overall profitability.

➤ 3.4. Daily Demand Analysis:

QUANTITY	COMPANY				
DAY	BISLERI	COCA-COLA	DABUR	PARLE AGRO	PEPSICO
Friday	45	344	44	81	78
Monday	113	402	40	85	82
Saturday	67	327	33	78	92
Tuesday	77	419	51	105	72
Wednesday	57	372	46	65	81
Grand Total	359	1864	214	414	405

Daily Demand Analysis

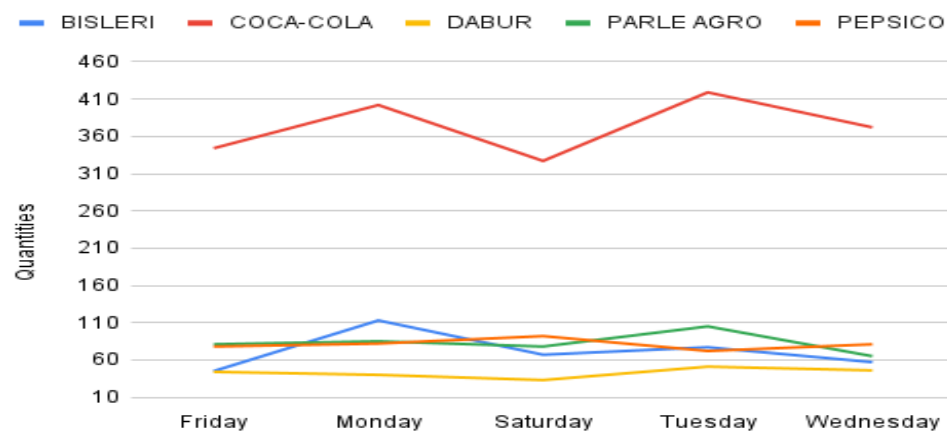


Figure:8

Daily demand analysis involves studying the demand for products or services on a daily basis to understand patterns, trends, and variations.

Key Findings:

Based on the analysis, Coca-Cola consistently shows higher demand compared to PepsiCo, Parle Agro, Bisleri, and Dabur across all days.

Dabur is observed to have the least demand among the mentioned brands in terms of quantities sold.

➤ 3.3. Monthly Demand Analysis:

A monthly demand analysis of products provides a comprehensive overview of the sales trends and patterns for each product over different months. By examining the monthly demand for products, businesses can identify seasonal variations, peak demand periods, and overall trends in customer preferences.

SUM of QUANTITY	MONTH		
COMPANY	August	July	September
BISLERI	71	240	48
COCA-COLA	683	751	430
DABUR	72	98	44
PARLE AGRO	110	196	108
PEPSICO	140	163	102
Grand Total	1076	1448	732

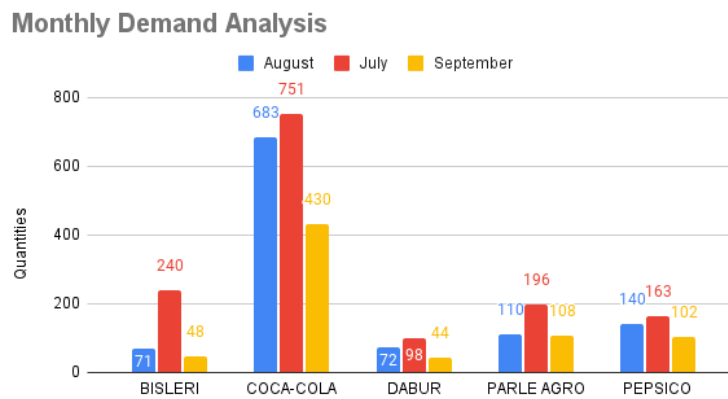


Figure:9:

Key Findings:

COCA-COLA is in more demand in all months followed by PepsiCo, Parle Agro, Bisleri. Dabur is least in terms of quantities.

➤ 3.4. Day Wise Analysis

Day-wise analysis involves examining sales or other relevant metrics on a daily basis to understand patterns, trends, and variations.

The daily sales figures for each category are as follows:

SUM of DAILY PROFIT	COMPANY				
DAY	BISLERI	COCA-COLA	DABUR	PARLE AGRO	PEPSICO
Friday	₹1,645.00	₹19,320.00	₹7,920.00	₹4,620.00	₹9,800.00
Monday	₹4,010.00	₹23,005.00	₹12,200.00	₹4,090.00	₹10,510.00
Saturday	₹2,450.00	₹19,120.00	₹5,940.00	₹3,477.00	₹12,640.00
Tuesday	₹2,725.00	₹25,040.00	₹8,880.00	₹5,165.00	₹8,870.00
Wednesday	₹2,220.00	₹19,665.00	₹8,280.00	₹3,565.00	₹9,990.00
Grand Total	₹13,050.00	₹106,150.00	₹43,220.00	₹20,917.00	₹51,810.00

Daily Profit Analysis

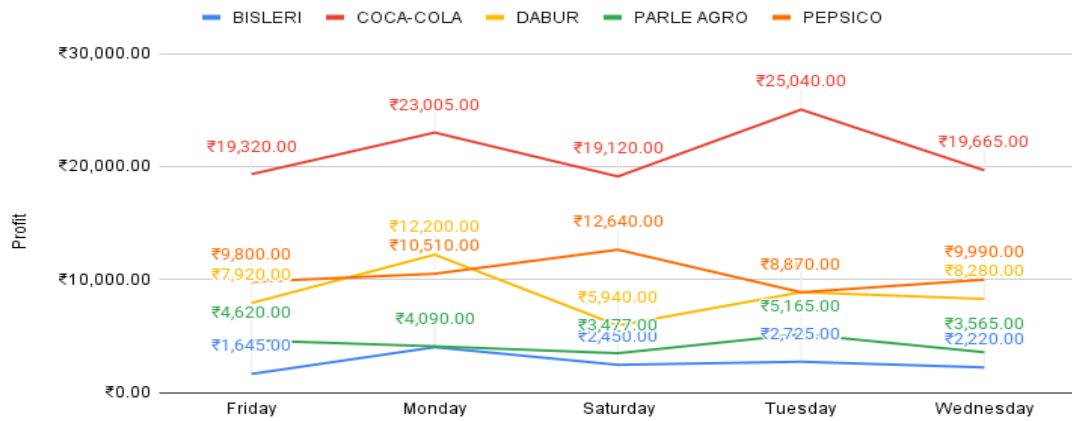


Figure:10:

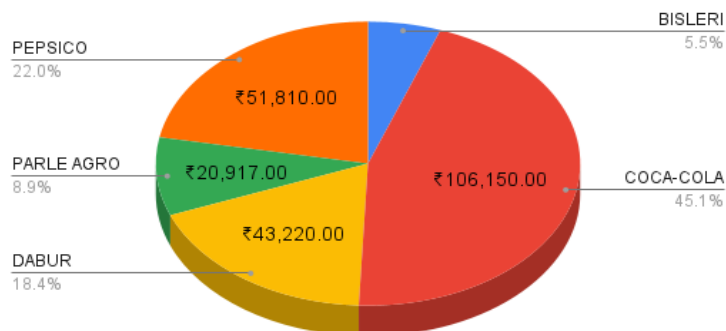
By creating chart, it became visually highlighted the daily variations in sales for each company's product throughout the week. This chart facilitates straight comparison across days and company, revealing any shifts or trends in sales volume.

➤ 3.5. Distribution Of Profit per Company

It illustrates how profits are allocated across different companies in a given dataset. Additionally, by understanding this distribution, owner can make strategic decisions regarding resource allocation, investment priorities, and area of improvement.

COMPANY	DAILY PROFIT
BISLERI	₹13,050.00
COCA-COLA	₹106,150.00
DABUR	₹43,220.00
PARLE AGRO	₹20,917.00
PEPSICO	₹51,810.00
Grand Total	₹235,147.00

Profit Distribution

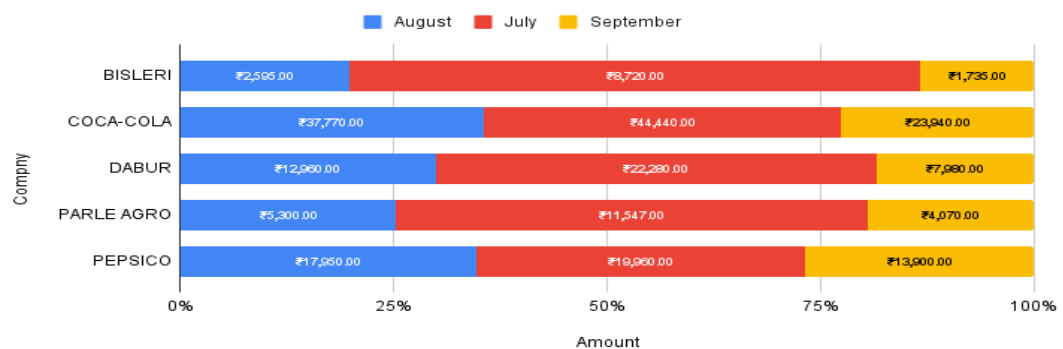


➤ 3.6. Month Wise Profit Analysis

The provided data comprises monthly sales data (in Indian Rupees) for five different Company's products: Bisleri, PepsiCo, Parle Agro, Dabur and Coca-Cola. The data covers a period of three months, spanning from July to September.

SUM of DAILY PROFIT	COMPANY				
MONTH	BISLERI	COCA-COLA	DABUR	PARLE AGRO	PEPSICO
August	₹2,595.00	₹37,770.00	₹12,960.00	₹5,300.00	₹17,950.00
July	₹8,720.00	₹44,440.00	₹22,280.00	₹11,547.00	₹19,960.00
September	₹1,735.00	₹23,940.00	₹7,980.00	₹4,070.00	₹13,900.00
Grand Total	₹13,050.00	₹106,150.00	₹43,220.00	₹20,917.00	₹51,810.00

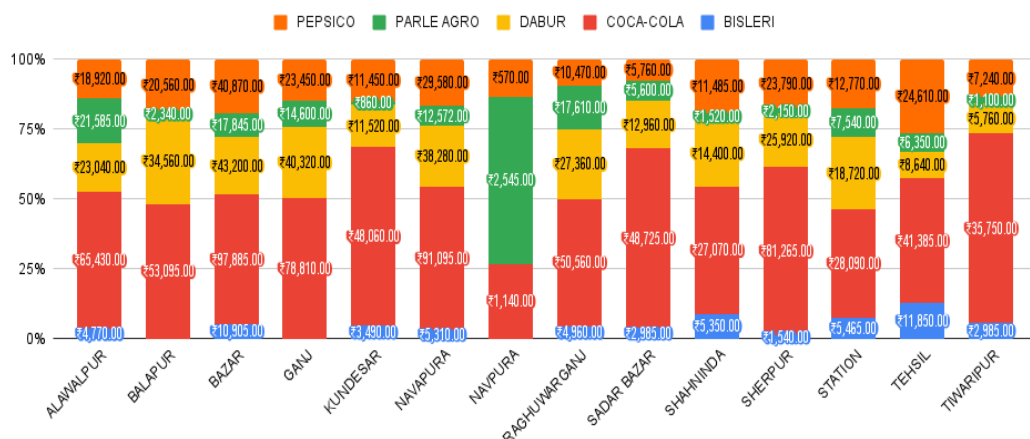
Monthly Profit Analysis



The vivid insights extracted from above chart creates an enlightening story about the distribution of total sales across the respective months. This graphical representation offers a overview of how sales were distributed over the months of July, August and September. The numerical breakdown is particularly striking: in July, profit claims 45.4% share of total, moving to August 32.5% of total, lastly September making up least 21.9% share of total.

➤ 3.7. Location Wise Sales and Profit Analysis

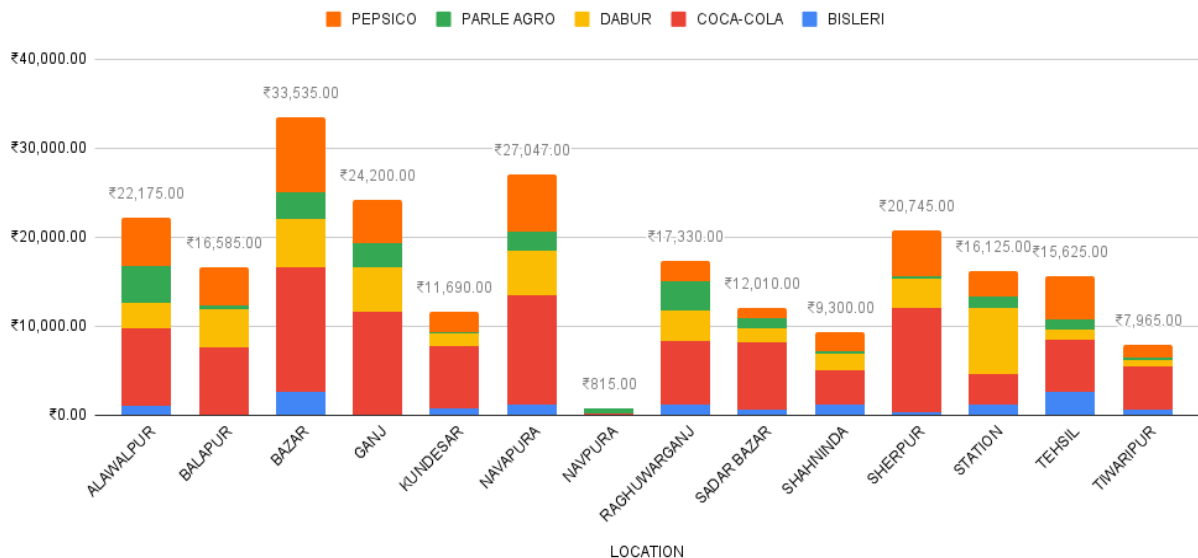
Location Wise Sales Analysis



SUM of TOTAL	COMPANY				
LOCATION	BISLERI	COCA-COLA	DABUR	PARLE AGRO	PEPSICO
ALAWALPUR	₹4,770.00	₹65,430.00	₹23,040.00	₹21,585.00	₹18,920.00
BALAPUR		₹53,095.00	₹34,560.00	₹2,340.00	₹20,560.00
BAZAR	₹10,905.00	₹97,885.00	₹43,200.00	₹17,845.00	₹40,870.00
GANJ		₹78,810.00	₹40,320.00	₹14,600.00	₹23,450.00
KUNDESAR	₹3,490.00	₹48,060.00	₹11,520.00	₹860.00	₹11,450.00
NAVAPURA	₹5,310.00	₹91,095.00	₹38,280.00	₹12,572.00	₹29,580.00
NAVPURA		₹1,140.00		₹2,545.00	₹570.00
RAGHUWARGANJ	₹4,960.00	₹50,560.00	₹27,360.00	₹17,610.00	₹10,470.00
SADAR BAZAR	₹2,985.00	₹48,725.00	₹12,960.00	₹5,600.00	₹5,760.00
SHAHNINDA	₹5,350.00	₹27,070.00	₹14,400.00	₹1,520.00	₹11,485.00
SHERPUR	₹1,540.00	₹81,265.00	₹25,920.00	₹2,150.00	₹23,790.00
STATION	₹5,465.00	₹28,090.00	₹18,720.00	₹7,540.00	₹12,770.00
TEHSIL	₹11,850.00	₹41,385.00	₹8,640.00	₹6,350.00	₹24,610.00
TIWARIPUR	₹2,985.00	₹35,750.00	₹5,760.00	₹1,100.00	₹7,240.00

The given data represents the sales figure of five different companies: Bisleri, Dabur, Coca-Cola, Parle Agro and PepsiCo, across various locations in Ghazipur, Uttar Pradesh. To visualise this 100% stacked column created, in this each location is represented by a column, and the height of each segment represents proportion of sales for each company.

Location Wise Profit Analysis



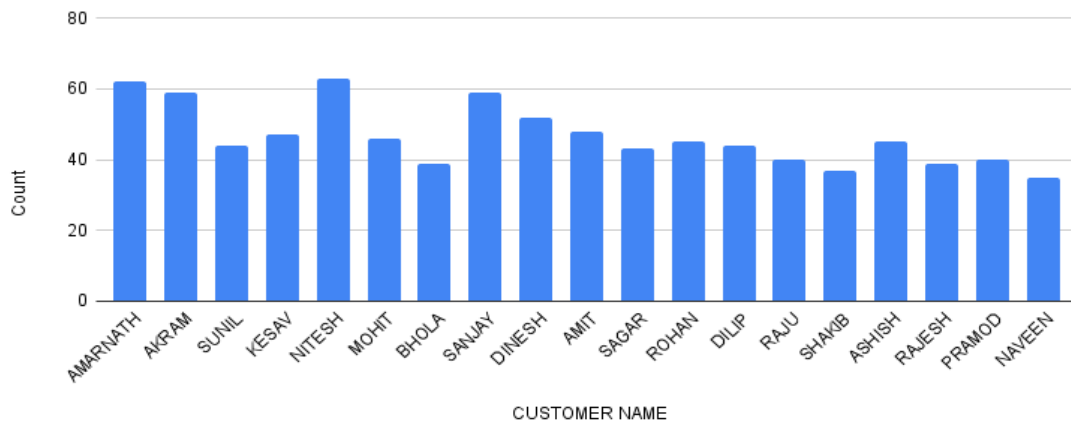
The above chart represents the profit figure across each location for different products of each company. To visualise this stacked column chart created, in this location represents by column, and the height of each segment represents proportion of profit for each company and total profit for each location is mentioned at top of each column.

Here, **Bazar** contributes more in sales as well as profit for each company followed by **Navapura** and **Alawalpur**. The least contributing location is **Navapura**.

➤ 3.8. Customer Analysis

The provided chart illustrates the count of customers associated with various names. A column chart has been generated to visually represents the customers.

Customer Analysis



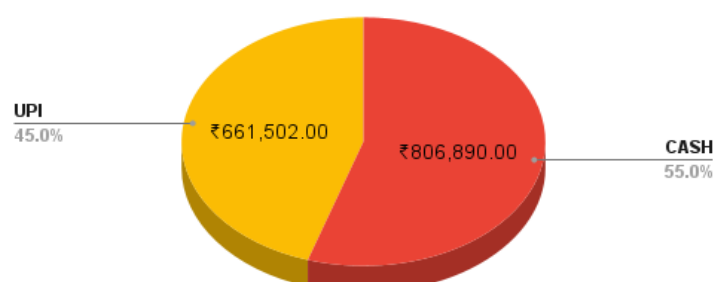
The height of each column shows the number of customers for each name. This visual representation accentuates the top customers and their respective customer counts(visits). It facilitates straightforward comparison among the various customers and aids in pinpointing the most noteworthy customer names based on their count.

Overall, the data indicates that Nitesh and Amarnath emerges as the top customers with the highest counts, followed by several others with substantial numbers.

➤ 3.9. Payment Method Analysis

Payment method analysis involves analysing the various methods customers use to make payments for services. This analysis aims to understand the preferences and effectiveness of different payment method used by customers. By analysing payment methods, business owner can gain insights about customer financial preferences, optimizing payment processes.

Payment Method



Key Analysis:

- Total sales: ₹1468392
- Total Cash Sale: ₹806890
- Total UPI Sale: ₹661502

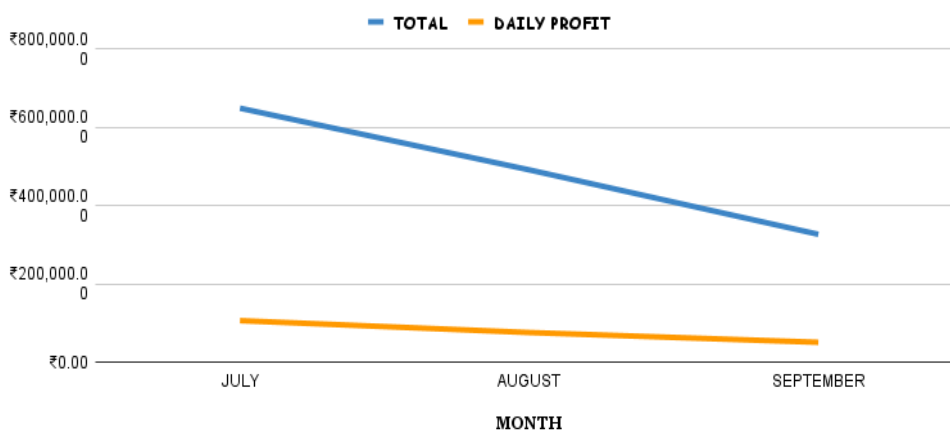
Additionally, a deeper analysis about the distribution of payments methods allows business to optimize financial operations. This could involve managing cash flows more effectively or improving online payment infrastructure to meet customer demands efficiently.

➤ 3.10. Revenue and Profit Analysis

To analyse the revenue and revenue and profit data on a monthly basis, it provided the following information:

MONTH	TOTAL	DAILY PROFIT
JULY	₹649,237.00	₹106,947.00
AUGUST	₹491,845.00	₹76,575.00
SEPTEMBER	₹327,310.00	₹51,625.00

TOTAL and DAILY PROFIT



July emerged as the top performer in monthly revenue, totalling ₹649,237.00, followed by August with ₹491,845. September recorded least monthly revenue of ₹327,310.00.

Regarding monthly profit, July led the pack with the highest profit of ₹106,947.00, followed by August with ₹76,575.00. September reported the lowest profit of ₹51,625.

4. Interpretation of Results and Recommendation

The analysis of the data obtained from SHIVA ENTERPRISES has provided valuable insights into the performance and trends of the business. Based on the findings, the following interpretations and recommendations can be made.

- **Demand Fluctuations:** The line chart (Fig 9) portraying the daily demand analysis reveals notable fluctuations in customer demand per company. Peaks on the chart coincide with periods of heightened demand, whereas valleys signify low-demand periods. Understanding these fluctuations holds crucial importance for resource allocation, and inventory management strategies.

Recommendations: Implement flexible inventory management strategies that can quickly adapt to changes in demand. During peak month and periods, ensure sufficient stock to meet customer needs, while optimizing resources and availability of some new and demanded items during slower periods to make consistent sales.

- **Inadequate financial management practices:** The assessment of financial management practices involved examining total sales, total profit, and total revenue for each location. Here's the breakdown of location wise analysis:

Bazar stands out with the highest total sales of ₹210705 and the highest profit of ₹33,535.00, indicating robust financial performance.

Navapura follows with 2nd highest total sales of ₹176,837 with total profit of ₹27,047. Despite having one of the lower total sales of ₹72,585 **Station** boasts the total profit of ₹16,125.00, indicating efficient financial management,

Recommendations: Based on this analysis, it's advisable to conduct further investigations into other locations to identify areas for improvement in financial management practises. This could involve analysing expenses, optimizing pricing strategies, or implementing cost-saving measures to enhance overall profitability across all locations.

- **Limited Market Penetration:** To address this, an analysis of daily and monthly sales figures across different companies and their product was conducted. Here's the summary of the findings:

July recorded the highest total sales and profit of ₹649,237 and ₹106,947 respectively.

Coca-Cola company and its products followed by Dabur and PepsiCo consistently displayed high sales across all months, indicating strong market demand. However, Bisleri exhibited relatively lower sales compared to other companies.

Recommendations: Based on this analysis, efforts should be directed towards replacing the product of Bisleri company with other alternative which is already in demand (such as Kinley by Coca-Cola). For least selling, strategies such as targeted marketing campaigns, exploring new customer segments, partnership campaign with local distributors and dealers can help to increase in sales. By focusing on improving the performance during least revenue months, business owner can capitalize on untapped market potential and enhance revenue.

- **Seasonal Sales Variations:** The analysis indicates that sales fluctuate throughout the year, peaking in July (specifically summer season) likely due to hot and humid seasons and other factors influencing consumer demand. While after August most of the sales generates from Fruit Drink items (Real) and Water because of marriage seasons and customer demand. Sometimes fluctuation occurs due to delay in Delivery citing weather conditions and other external reasons, it also effects on overall revenue.

Recommendations: To make out from the most of these peak periods, it's advisable for the business to plan promotions strategically, increase in capacity of the products that are in demand during least revenue generating months, and optimize its distribution network to meet the surge in demand effectively.

- **Innovative pricing approaches to stay competitive:** It involves setting prices for products or services based on competitor pricing, market conditions, and customer perceptions. Every market contains lot of competitors and challenges and to became consistent in terms of tackling them is very necessary.

Recommendations: The goal is to set prices that are competitive yet profitable, taking into account factors such as transportation costs, brand positioning, and the value proposition offered to customers. By conducting through market analysis encompassing competitor pricing, customer preferences, and market trends business can develop competitive pricing strategies, meet the evolving needs of customers, increase market share, and maintain profitability in a competitive marketplace.

➤ **CONCLUSION:**

In conclusion, the analysis of data has provided valuable insights into the performance of SHIVA ENTERPRISES. By examining key metrics such as quantities sold, total sales, and daily profits for each product category, the business can identify trends, capitalize on peak sales periods, and optimize its operations. The analysis also revealed clear seasonal variations in sales, highlighting the importance of strategic planning and resource allocation during peak periods. Furthermore, the recommendation to establish a data-driven decision-making culture will enable the business to continuously improve its strategies, better understand customer preferences, and enhance its competitive position in the market. Regular monitoring of performance and the impact of implemented solutions is crucial for sustained success. Continuous analysis of financial metrics, pricing strategies, and market penetration efforts helps identify areas for improvement and adapt recommendations accordingly. Overall, by leveraging data-driven insights and implementing the recommended strategies, SHIVA ENTERPRISES can achieve sustainable growth and long-term success.